

Rotationally symmetric Ricci flow

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Introduction to Ricci Flow

Definition (Ricci flow)

Given a family of Riemannian metrics $(\mathcal{M}^n, g(t))_{t \in [0, T]}$. $(\mathcal{M}^n, g(t))_{t \in [0, T]}$ is called a *solution of Ricci flow* (or shortly, a *Ricci flow*) if it satisfies

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on $\mathcal{M} \times [0, T]$.

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- [Hamilton, '82] **Existence** and **uniqueness** of RF for the compact case.
- [Deturck, '83] Simply the proof and introduced *Ricci-Deturck flow*.
- Non-compact manifolds only have partial progress.

Existence theorem of complete Ricci Flow

A non-exhaustive list of the existence of Ricci flow.

- [Shi, '89] Complete and $|\text{Rm}|_\infty < \infty$.

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- [Simon, '02] If the initial metric g is $\mathcal{C}^0$ -closed to a smooth metric h with $|\mathrm{Rm}(h)|_\infty < \infty$, that is,

$$(1 - \epsilon(h))h \leq g \leq (1 + \epsilon(h))h, \quad (2)$$

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- [Cabezas-Rivas-Wilking, '15] **Complete** and satisfies

$$\mathrm{Rm}(g) + \mathcal{I} \in \mathcal{C}_{\mathrm{PIC}2} \Rightarrow \sec(g) \geq -1, \quad (3)$$

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then $\exists$ a complete RF starting with $(\mathcal{M}^n, g)$.

Remark: Lack of curvature bounds for positive time if we don't impose volume control.

Existence theorem of complete Ricci Flow

- [Lai, '19] **Complete** and satisfies the following:

$$\begin{cases} \text{Vol}_g(x, 1) \geq v_0 > 0 & \text{for all } x \in \mathcal{M}^n, \\ \text{Rm}(g) + \mathcal{I} \in \mathcal{C}_{\text{PIC1}} \Rightarrow \text{Ric}(g) \geq -(n-1), \end{cases}$$

$\exists$ a complete RF $(\mathcal{M}^n, g(t))_{t \in [0, T]}$ starting with $(\mathcal{M}^n, g)$ s.t.

$$\begin{cases} |\text{Rm}(x, t)| \leq \frac{c}{t}, \\ \text{inj}_{g(t)}(x) \geq \sqrt{\frac{t}{c}}, \end{cases} \quad (4)$$

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- [Chan-Chau-Chen-Huang-Lee-Lee-Martens, etal] **Complete** and satisfies

$$\begin{cases} \text{Vol}(x, r) \leq v_0 r^n \text{ for all } (x, r) \in \mathcal{M}^n \times (0, \infty); \\ \text{Uniform local Sobolev constant } C_S; \\ \|\text{Rm}\|_{\mathcal{L}^{\frac{n}{2}}(\mathcal{M}^n)} \ll \varepsilon(n, C_S, v_0), \end{cases} \quad (5)$$

then $\exists$ a complete *long-time* RF starting with $(\mathcal{M}^n, g)$ which satisfies (4).

Introduction to RSRF

- $(\mathbb{R}^{n+1}, g_0 = dx^2 + f(x)^2 g_{\text{std}})$: complete and rotationally symmetric (shortly, RS), where g_{std} : the standard metric on S^n .

Then *the smoothness at o* is equivalent to for all $k \geq 0$,

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Not easy to solve this PDE in general with a boundary condition.

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Existence using Lai's result:

- $\text{Rm}(g_0) + \mathcal{I} \in \mathcal{C}_{\text{PIC1}} \Rightarrow 1 + \frac{1-(f_x)^2}{f^2} \geq 0 \Rightarrow f(x) = O(e^x).$

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Motivation: Could we resolve the singular RS manifold, e.g., singular at o , by an *instantaneously complete RSRF*?

Main Result - Existence theorem of RSRF

Theorem (Hsiao, '26)

Let $(\mathbb{R}^{n+1}, g := dx^2 + f(x)^2 g_{std})$ be as above. Suppose that $\liminf_{x \rightarrow \infty} f(x) > 0$. Then $\exists$ a complete RSRF $(\mathbb{R}^{n+1}, g(t))_{t \in [0, T_g]}$ with $g(0) = g$ and a constant $\Lambda_g > 0$ s.t.

$$|\text{Rm}(g(t))| \leq \frac{\Lambda_g}{t}, \quad (9)$$

on $\mathbb{R}^{n+1} \times (0, T_g]$.

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$$\begin{cases} |\text{Rm}(g)| \leq \ell^{-2}, & \text{on } B_g(o, 6\ell); \\ f_x \geq \delta, & \text{on } B_g(o, 3\ell); \\ f \geq \varepsilon\ell, & \text{on } \mathbb{R}^{n+1} \setminus B_g(o, \frac{1}{2}\ell). \end{cases} \quad (10)$$

Then $\Lambda_g = \Lambda(n, \delta, \varepsilon) > 0$ and $T_g = T(n, \delta, \varepsilon)\ell^2 > 0$.

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We point out that such solutions, complete RF with scaling invariant curvature estimate (9), are unique [Lee, '25].

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Remark: Two-sided curvature bounds are too restricted.

Main Result - Improving curvature estimate when $f_x \geq 0$

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on $\mathbb{R}^{n+1} \times (0, T_g]$. Furthermore, suppose that $\ell, \delta > 0$ are two constants s.t.

$$\mathcal{R}(g) \geq -\ell^{-2} \text{ and } f_x \geq \delta, \quad (12)$$

on $B_g(o, 4\ell)$. Then $\Lambda_g = \Lambda(n, \delta) > 0$ and $T_g = T(n, \delta)\ell^2 > 0$.

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As a consequence, if $\mathcal{R}(g) \geq 0$ and $f_x \geq \delta > 0$ on $\mathbb{R}^{n+1}$, then the flow exists for all $t \geq 0$ with (11) by taking $\ell = \infty$.

Main result - Smoothing conical singularity by RSRF

One application is to smooth out RS manifolds w/ conical singularity at o .

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Theorem (Hsiao, '26)

Let $((0, \infty) \times S^n, g := ds^2 + f(s)^2 g_{std})$ be an incomplete manifold **with uniform Ricci lower bound**. Suppose that

- $f_s \geq 0$ for all $s > 0$.
- $f_s(0) \in (0, 1)$ and $f^{(k)}(0) = 0$ for all $k \neq 1$.

Then there $\exists$ two constants $\hat{\Lambda}(n, g)$, $\hat{T}(n, g) > 0$ and a complete RSRF $(\mathbb{R}^{n+1}, g(t))_{t \in (0, \hat{T}]}$ such that

$$|\text{Rm}(g(t))| \leq \frac{\hat{\Lambda}}{t}$$

on $\mathbb{R}^{n+1} \times (0, \hat{T}]$, and $g(t) \xrightarrow{C_{loc}^\infty(\mathbb{R}^{n+1} \setminus \{o\})} g$ as $t \rightarrow 0$.

Moreover, $(\mathbb{R}^{n+1}, d_{g(t)}, o) \xrightarrow{pGH} (\mathbb{R}^{n+1}, d_g, o)$ as $t \rightarrow 0$.

Idea of the proof

We approximated g by the following wrapped functions:

$$f_k(x) := \phi(x - k)f(x) + (1 - \phi(x - k))\frac{1}{2}\varepsilon\ell, \quad (13)$$

where $\phi : \mathbb{R} \rightarrow [0, 1]$ is a fixed smooth cut-off function with $\phi \equiv 1$ on $(-\infty, 0]$ and $\phi \equiv 0$ on $[1, \infty)$. Then $g_k \rightarrow g$ smoothly as $k \rightarrow \infty$ and g_k is cylindrical as $x > k + 1$. $\Rightarrow g_k$ **has bounded geometry**.

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Idea: Run complete RSRF $(\mathbb{R}^{n+1}, g_k(t))_{t \in [0, \tilde{T}_k]}$ on each $(\mathbb{R}^{n+1}, g_k)$, extend each flow to a uniform time $\hat{T} > \tilde{T}_k$ with

$$|\text{Rm}(g_k(t))| \leq \frac{\hat{\Lambda}}{t}, \quad (14)$$

on $\mathbb{R}^{n+1} \times (0, \hat{T}]$ for all k .

Then we can take a subsequential limit as the solution.

Key ingredients in the main theorem

Pseudolocality theorem: If for all $k \geq 1$, $r \in (0, 1]$,
 $(x, t) \in \mathbb{R}^{n+1} \times [0, \tilde{T}_k]$,

$$\frac{\text{Vol}_{g_k(t)}(x, r)}{r^{n+1}} \geq \nu, \quad (15)$$

then $\exists \hat{\Lambda}_1(n, \nu)$, $\hat{T}_1(n, \nu) > 0$ such that

$$|\text{Rm}(g_k(t))| \leq \frac{\hat{\Lambda}_1}{t}, \quad (16)$$

on $\mathbb{R}^{n+1} \times (0, \tilde{T}_k \wedge \hat{T}_1]$.

Key ingredients in the main theorem

Pseudolocality theorem: If for all $k \geq 1$, $r \in (0, 1]$,
 $(x, t) \in \mathbb{R}^{n+1} \times [0, \tilde{T}_k]$,

$$\frac{\text{Vol}_{g_k(t)}(x, r)}{r^{n+1}} \geq \nu, \quad (15)$$

then $\exists \hat{\Lambda}_1(n, \nu)$, $\hat{T}_1(n, \nu) > 0$ such that

$$|\text{Rm}(g_k(t))| \leq \frac{\hat{\Lambda}_1}{t}, \quad (16)$$

on $\mathbb{R}^{n+1} \times (0, \tilde{T}_k \wedge \hat{T}_1]$.

Sketch: Argue by contradiction. Perelman's point-picking method implies a non-trivial *ancient RSRF* $(\mathbb{R}^{n+1}, h(t))_{t < 0}$ with $|\text{Rm}| < \infty$ and $AVR > 0$. Thanks to the *generalized Hamilton-Ivey estimate* [Zhang, '15], we obtain $\text{Rm}(h(t)) \geq 0$ for all $t < 0$. This contradicts $AVR > 0$. $\square$

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Question: How to control the volume ratio along RF?

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Lemma

Let $(\mathbb{R}^{n+1}, g := dx^2 + f(x)^2 g_{std})$ be a complete RS manifold. Suppose that $\exists \delta > 0$ such that

$$f(x) \geq \delta x \chi_{[0,1]}(x) + \delta \chi_{(1,\infty)}(x), \quad (17)$$

for all $x \geq 0$. Then there $\exists v(\delta, n) > 0$ such that

$$\frac{\text{Vol}_g(B_g(p, r))}{r^{n+1}} \geq v, \quad (18)$$

for all $p \in \mathbb{R}^{n+1}$ and $r \in (0, 1]$.

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It remains to prove $f_{s_t} \geq \delta$ on $B_{g_k(t)}(o, 1)$ and $f \geq \delta$ on $B_{g_k(t)}(o, 1)^c$ for some uniform δ (only depending on $g(0)$).

A priori estimate of f along RF

Lemma

Let $(\mathbb{R}^{n+1}, g(t) := ds_t^2 + f(s_t, t)^2 g_{std})_{t \in [0, T]}$ be defined as before. Furthermore, assume $|\text{Rm}(g(t))| \leq \alpha/t$ on $\mathbb{R}^{n+1} \times [0, T]$ and at $t = 0$, there exist some constants $w, \delta, \varepsilon > 0$ such that

$$\begin{cases} |\text{Rm}(g(0))| \leq \ell^{-2}, & \text{on } B_0(o, 6\ell); \\ |\text{Rm}(g(0))| \leq (w\ell)^{-2}, & \text{outside } B_0(o, M\ell); \\ f_s(s, 0) \geq \delta, & \text{on } B_0(o, 3\ell); \\ f(s, 0) \geq \varepsilon\ell, & \text{when } s \geq \frac{1}{2}\ell. \end{cases} \quad (19)$$

Then $\exists v_1(n, \delta, \varepsilon), \hat{T}_2(n, \alpha, \delta, \varepsilon, w) > 0$ such that

$$\frac{\text{Vol}_{g(t)} B_{g(t)}(x, r)}{r^{n+1}} \geq v_1, \quad (20)$$

on $\mathbb{R}^{n+1} \times [0, T \wedge \hat{T}_2\ell^2]$ and for all $r \in (0, \ell]$.

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It's crucial that v_1 is independent of α .

Ricci flow with scaling-invariant estimate

Local estimates under $|\mathrm{Rm}(x, t)| \leq \alpha/t$ decay:

- ① [Perelman, '02] $d_t(x, y) - c_1(n)\sqrt{\alpha t} \geq d_s(x, y) - c_1(n)\sqrt{\alpha s}$ on $0 \leq s \leq t \leq T$.

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- ③ [Simon-Topping, '16] If $\mathcal{R}(0) \geq -K$ on $B_0(p, 1)$, then $\exists T_1(n, \alpha, K) > 0$ s.t. $\mathcal{R}(p, t) \geq -2K$ for $0 \leq t \leq T \wedge T_1$.

$$\square \mathcal{R} := \left(\frac{\partial}{\partial t} - \Delta_{g(t)} \right) = 2|\mathrm{Ric}|^2 \geq \frac{2\mathcal{R}^2}{n}. \quad (21)$$

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- ④ [Lee-Tam, '20] Suppose that $L, \varphi \in C^0(B_0(p, 2) \times [0, T])$ with $\varphi, L \leq \alpha/t$. If

$$\square \varphi \leq L\varphi \text{ and } \varphi(x, 0) \leq 0, \quad (22)$$

then $\exists T_2(n, \alpha)$ s.t. $\varphi(p, t) \leq t^{\alpha+2}$ on $[0, T \wedge T_2]$.

A priori estimate of f along RF

Set $\ell = 1$ and $T \gg 1$. The evolution equation of f_{s_t} is

$$\square f_{s_t} := \left(\frac{\partial}{\partial t} - \Delta_{g(t)} \right) f_{s_t} = \frac{\mathcal{R}(g(t))}{n} f_{s_t}. \quad (23)$$

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- ⑤ Consider $\phi(x, t) := \delta(1 - 2n(n+1)t) - f_{s_t}$, then

$$\square \phi = \frac{\mathcal{R}}{n} \phi - \frac{\mathcal{R}}{n} \delta(1 - 2n(n+1)t) - 2n(n+1)\delta \leq \frac{\mathcal{R}}{n} \phi,$$

on $B_0(o, 5) \times [0, T_3 \wedge \frac{1}{4n(n+1)}]$.

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- ⑥ Lee-Tam's lemma: $\phi \leq \delta/4$ on $B_0(o, 3) \times [0, T_4(n, \alpha, \delta)]$. Hence, $f_{s_t} \geq \frac{\delta}{2} - \frac{\delta}{4} = \frac{\delta}{4}$ on $B_0(o, 3) \times [0, T_4]$.

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- ⑦ Step 2. implies $f_{s_t} \geq \frac{\delta}{4}$ on $B_t(o, 2)$ and $t \in [0, T_5]$.

A priori estimate of f along RF

Now we remain to show $f \geq \frac{\delta}{4}$ on $B_t(o, 1)^c$. The eq of f can be written as

$$\left(\frac{\partial}{\partial t} - \frac{\partial^2}{\partial s_t^2} \right) \left[e^{(n-1)(n-2)t} \left(f^n + \frac{2}{n-2} \right) \right] \geq 0. \quad (24)$$

Apply minimum principle on $s_0^{-1}([\frac{1}{2}, M + 2w]) \times [0, T_5]$ and let $M \rightarrow \infty$.

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After shrinking T_6 again if necessary, we get $f \geq v_1(n, \delta, \varepsilon)$ on $B_0(o, 1/2)^c \times [0, T_7(n, \delta, \varepsilon, w, \alpha)]$. □

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Finally, we use these two lemmas to do bootstrapping argument to extend each $(\mathbb{R}^{n+1}, g_k(t))_{t \in [0, \hat{t}]}$.

Remarks on smoothing result

- In the case of $f_x(x, 0) \geq 0$, the approximation f_k is a *straight line with positive slope* outside $[0, k + 1]$. Hence, we automatically have $(f_k)_s(s, t) \geq 0$ by standard maximum principle on **complete RF w/ bounded curvature**. Thus, we remove the assumptions of upper bounds on $|\text{Rm}(g(0))|$ near o and the lower bound of f at ∞ .

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- On complete RSRF with $|\text{Rm}(x, t)| \leq \alpha/t$, $\text{Ric}(g(0)) \geq -K$ implies $\text{Ric}(g(t)) \geq -K'(n, K, \alpha, T)$ on $\mathbb{R}^{n+1} \times [0, T]$.

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- For complete RF $(\mathcal{M}^n, g(t))_{t \in (0, T)}$ with $|\text{Rm}(x, t)| \leq \alpha/t$ and $\text{Ric}(x, t) \geq -1$, there $\exists$ a complete metric space $(\mathcal{M}^n, d_0)$ such that

$$(\mathcal{M}^n, d_{g(t)}, p) \xrightarrow{\text{pGH}} (\mathcal{M}^n, d_0, p) \text{ as } t \rightarrow 0. \quad (25)$$

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- For singular RS metric, we do the approximation carefully. $f^k(0) = 0$ for $k \geq 3$ is needed in our construction.

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- In the case of $f_x(x, 0) \geq 0$, the approximation f_k is a *straight line with positive slope* outside $[0, k + 1]$. Hence, we automatically have $(f_k)_s(s, t) \geq 0$ by standard maximum principle on **complete RF w/ bounded curvature**. Thus, we remove the assumptions of upper bounds on $|\text{Rm}(g(0))|$ near o and the lower bound of f at ∞ .
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- Ricci lower bounds ensure the pointed Gromov-Hausdorff convergence at $t = 0$.
- The boundary condition $f^k(0) = 0$ for $k \geq 3$ could probably be removed.

Thanks for your listening!